CLASS – TYBBA CA , TY BCA

ASSIGNMENT – 4

SUBJECT – ESSENTIALS OF CLOUD COMPUTING

Q1.Define Cloud Security. What is the Shared Responsibility Model? Who is responsible for what - Provider vs Customer?

Q2. Explain Identity and Access Management (IAM) and Multi-Factor Authentication (MFA). Differentiate between Authentication and Authorization.

Q3. What is Data Encryption in cloud computing? Explain encryption at rest, in transit and in use.

Q4. What is GDPR? Mention its objectives and compare GDPR with Information Technology (IT) Act, 2000 of India.

SECTION (D) - LONG ANSWER TYPE QUESTIONS - 10 Marks Each - For Q.3 & Q.4 in University Paper - MUST PREPARE

Q5.* Explain Cloud Security Fundamentals in detail. Discuss the Shared Responsibility Model in IaaS, PaaS, SaaS and explain best security practices like MFA, Least Privilege, Zero Trust.

Q6. Describe Identity and Access Management (IAM) in detail. Explain its components, principles - Principle of Least Privilege, advantages, disadvantages and implementation in AWS/Azure.

Q7. Explain Data Encryption in Cloud Computing in detail. Discuss symmetric vs asymmetric encryption - AES-256, RSA - their advantages, disadvantages, and Key Management models - BYOK, HYOK, Cloud KMS.

Q8. Explain Cloud Security Threats and Risks in detail. Discuss various threats such as cloud misconfiguration due to improperly configured S3 buckets, data breaches, insecure APIs, ransomware, insider threats, account hijacking, and DDoS attacks with real-world examples.

Q9. What is a Service Level Agreement (SLA)? Explain its types, key metrics - RTO, RPO, MTTR, MTBF, benefits, challenges, and best practices. Why is SLA important for cloud compliance?

Q10. Explain the emerging trends in cloud computing including Serverless Computing, Edge Computing, Fog Computing, AI and Machine Learning Cloud Environment, and their applications. Differentiate between Edge vs Fog.

Q11. What is a Multi-Cloud Strategy? Explain its characteristics, advantages, challenges, reasons why organizations adopt multi-cloud environments - Avoiding Vendor Lock-in, Best-of-Breed Services, Resilience, Cost Optimization. Give real-world example - Bank using AWS, Azure, GCP.

Q12. Discuss career opportunities in Cloud Computing. Explain the roles and responsibilities of Cloud Engineer, Cloud Architect, DevOps Engineer, Cloud Security Engineer, IAM Specialist, Data Engineer, ML Engineer, and FinOps Analyst along with common certifications path - AWS, Azure, GCP.

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